Custom Rescue Node Pinout Guide

> **Note:** This guide maps the exact 11 GPIO pins available on the **Seeed Studio XIAO ESP32C3** to the SX1262 LoRa, NEO-6M GPS, and MPU6050 components. This uses 100% of the board's available IO.

Overall Pin Assignment Summary

XIAO Pin	ESP32-C3 GPIO	Component	Function
D8	`GPIO 8`	**SX1262 LoRa**	SPI SCK (Clock)
D9	`GPIO 9`	**SX1262 LoRa**	SPI MISO
D10	`GPIO 10`	**SX1262 LoRa**	SPI MOSI
D3	`GPIO 5`	**SX1262 LoRa**	SPI NSS (Chip Select)
D1	`GPIO 3`	**SX1262 LoRa**	DIO1 (Interrupt)
D2	`GPIO 4`	**SX1262 LoRa**	RESET
D0	`GPIO 2`	**SX1262 LoRa**	BUSY
D7	`GPIO 20`	**NEO-6M GPS**	TX (Connects to GPS TX)
D6	`GPIO 21`	**NEO-6M GPS**	RX (Connects to GPS RX)
D4	`GPIO 6`	**MPU6050**	I2C SDA
D5	`GPIO 7`	**MPU6050**	I2C SCL

> **Warning:** Because this layout consumes all 11 pins, you will not have spare pins for external physical buttons or indicator LEDs. The built-in boot button on the XIAO

> board will act as your main Meshtastic user button.

Power Connections

XIAO Pin	Connects To
3V3	SX1262 VCC, GPS VCC, MPU6050 VCC
GND	SX1262 GND, GPS GND, MPU6050 GND